

Part 2 — From Inference to Discovery (Modules 5–7)

INFO 521 · Machine Learning Foundations · project overview

How to read this

This overview is the project brief. The milestone notebooks carry the authoritative Satisfactory criteria, each stated at the top of its own notebook. Grading mechanics, including the gate and bundle structure, are on the course Assessment page.

Course-schedule placement. Part 2 of the project: begins at the midpoint (Week 5), due Week 7.5. Milestones on the module arc: 2.1 Approximate inference (M5, due Wk 6), then 2.2 Classification and SVM (M6, due Wk 6), then 2.3 Clustering (M6, due Wk 7), then 2.4 PCA (M7, due Wk 7), then 2.5 Synthesis (due Wk 7.5). Gating: Checkpoint quiz 2.1 (Gate A, Wk 5), quiz 2.3 (Gate B, Wk 6).

The second half of the course project, beginning at the midpoint, due Week 7.5.

Part 2 continues the *same* clinical cohort from Part 1 and extends it from prediction into discovery. Two things expand at once: the **method** (regression → classification → unsupervised) and the **feature space** (single predictor → the six-feature non-BP set). It is organized as **two thematic gates plus a capstone**.

Milestones

Gate A — Supervised (predict the hypertension label from the non-BP features)

1. **2.1 Approximate Bayesian inference** (M5) — Bayesian logistic regression; from-scratch log-posterior / gradient / Hessian (gradient-checked), Newton–Raphson (MAP), Laplace approximation, Metropolis sampler; MAP/Laplace/Metropolis mapped to P1’s point/likelihood/posterior. Gated by **Checkpoint quiz 2.1**.
2. **2.2 Classification** (M6) — from-scratch soft-margin SVM (primal hinge + L2, Pegasos), precision/recall/F1, the dual + kernel trick (conceptual), probabilistic-vs-margin showdown on one split.

Gate B — Unsupervised (drop the labels; find structure)

3. **2.3 Clustering** (M6) — from-scratch k-means, random restarts for local minima, K selection (elbow / silhouette), and the cluster-vs-held-out-label reveal. Gated by **Checkpoint quiz 2.3**.
4. **2.4 Dimensionality reduction** (M7) — from-scratch PCA (covariance/SVD), variance explained, biplot, and the k-means clusters re-viewed in PC space.

Capstone

5. **2.5 Synthesis** — one narrative across the whole semester on one cohort; supervised vs. unsupervised argument; structured peer review.

The additive bridge

Part 1 ended on a closed-form conjugate Gaussian posterior. Part 2 breaks it deliberately:

1. **Approximate Bayesian inference (M5).** Binarize systolic BP into a **hypertension** label (ACC/AHA: $SBP \geq 130$ or $DBP \geq 80$) and predict it from the **non-BP** features — the blood-pressure columns are excluded so the label can't leak into its own predictors. A Bayesian **logistic regression** model has **no conjugate posterior** — so you implement, from scratch: Newton–Raphson (MAP), the Laplace approximation, and a Metropolis sampler. Compare what each buys you.
2. **Classification (M6).** The same binary label is now an SVM target. Soft-margin primal (with the dual and kernel trick derived conceptually). Evaluate with precision / recall / F1. Contrast the discriminative SVM with the probabilistic classifier from step 1.
3. **Clustering (M6).** Drop the labels. Implement k-means from scratch on the six-feature set; address local minima (restarts) and model selection (choosing K). Do the discovered clusters relate to the hypertension label?
4. **Dimensionality reduction (M7).** PCA from scratch on the features; visualize the cohort in 2-D; feed components into the clustering and discuss.

Gates & bundle (specs)

The Part 2 bundle is **Satisfactory** when:

- **Gate A — Supervised:** 2.1 and 2.2 Satisfactory, and **Checkpoint 2.1** (Newton–Raphson update) passed.
- **Gate B — Unsupervised:** 2.3 and 2.4 Satisfactory, and **Checkpoint 2.3** (k-means objective / PCA) passed.
- **Capstone:** 2.5 synthesis Satisfactory (structured peer review completed).

See the Assessment page for the specs mechanics. The checkpoint quizzes that gate these milestones run on D2L; each is due the week before its milestone.

Play, extended

- A held-out **prediction round** on the binary outcome (compare SVM vs. Bayesian logistic regression on the same split).
- A **cluster-vs-label reveal**: do unsupervised subgroups recover clinical hypertension strata you never showed the algorithm?
- PCA **biplot exploration**: which features drive the principal axes?

Carry-over outcomes

The same plumbing applies: `info521.data.hypertension(ds)` provides the leakage-safe Gate A label (ACC/AHA hypertension); `load_clinical()` returns the feature matrix with the blood-pressure columns (`sbp`, `dbp`) excluded, so the Gate A/B feature set is non-BP by construction; `checks.expect_gradient` verifies the 2.1 gradient. No estimators are added to the library — k-means, PCA, SVM, and the samplers are all student-implemented.